**Case Study 5: Password Policy Validation**

**1. Problem statement**

Develop a Python program to validate a password entered by the user. The program should check whether the password satisfies basic security requirements, such as minimum length, uppercase letters, lowercase letters, digits, and special characters. It should display whether the password is valid or invalid.

**2. Objectives**

- To accept a password from the user.
- To check password length and character types.
- To identify whether the password meets the required rules.
- To practice strings, loops, conditional statements, and variables in Python.

**3. Assumptions**

For this program, we will use the following sample rules:

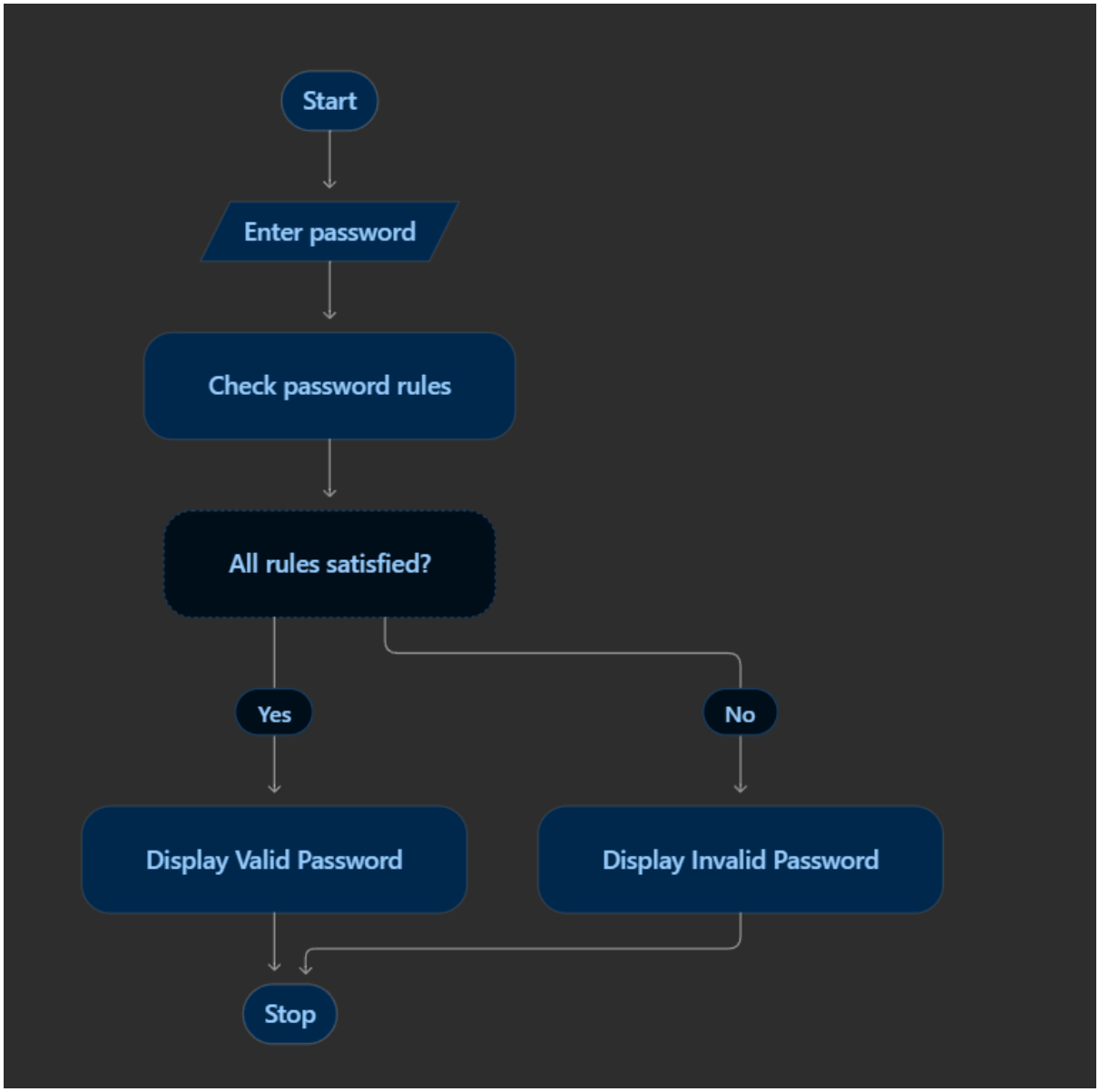
- The password must contain at least 8 characters.
- It must contain at least one uppercase letter.
- It must contain at least one lowercase letter.
- It must contain at least one digit.
- It must contain at least one special character, such as @, #, or !.
- The password is considered valid only when all five rules are satisfied.

**4. Algorithm**

1. Start the program.
2. Accept a password from the user.
3. Initialize variables to check uppercase letters, lowercase letters, digits, and special characters.

4. Check every character in the password.
5. Check whether the password has at least 8 characters.
6. If all requirements are satisfied, display Valid Password.
7. Otherwise, display Invalid Password.
8. Stop the program.

**5. Flowchart**



**6. Pseudocode**

```
START

INPUT password

SET uppercase = false
SET lowercase = false
SET digit = false
SET special = false

FOR each character in password
  IF character is uppercase
    SET uppercase = true
  ELSE IF character is lowercase
    SET lowercase = true
  ELSE IF character is a digit
    SET digit = true
  ELSE
    SET special = true
  END IF
END FOR

IF length of password >= 8 AND
  uppercase AND lowercase AND digit AND special
```

```
   DISPLAY "Valid Password"
ELSE
   DISPLAY "Invalid Password"
END IF

STOP
```

**7. Source code**

```
# Password Policy Validation

password = input("Enter your password: ")

uppercase = False
lowercase = False
digit = False
special = False

for ch in password:
   if ch.isupper():
      uppercase = True
   elif ch.islower():
      lowercase = True
   elif ch.isdigit():
      digit = True
   else:
```

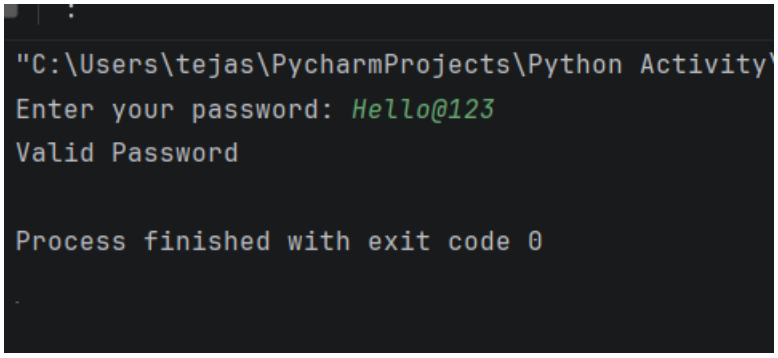
```
    special = True

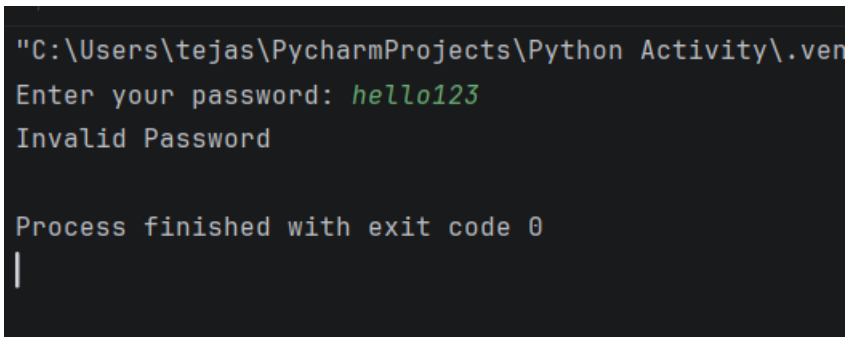
if (len(password) >= 8 and uppercase and lowercase
    and digit and special):
  print("Valid Password")
else:
  print("Invalid Password")
```

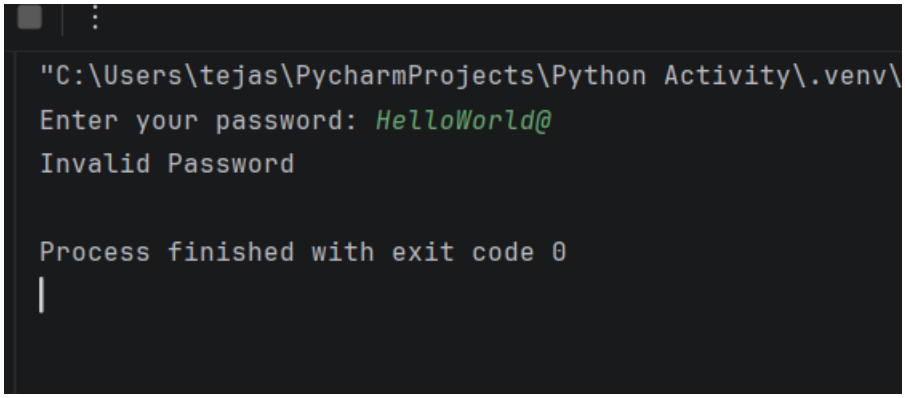
**8.Test case**

| Password entered | Expected output |
|---|---|
| Hello@123 | Valid Password |
| hello123 | Invalid Password |
| HelloWorld@ | Invalid Password |

**9.Output**






## 10. Conclusion

**The Password Policy Validation program checks whether a password satisfies basic security requirements. It uses strings, loops, conditional statements, and built-in character-checking methods to validate the password entered by the user.**